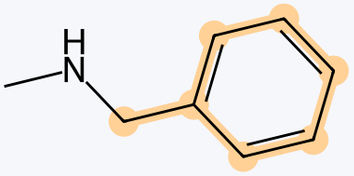
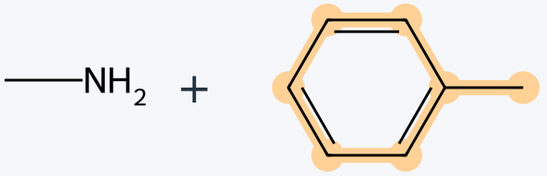
a Greedy constraints and the single-edge sweep

Base



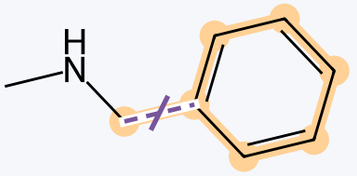
Large benzyl region highlighted

Target



A large region can be preserved

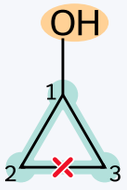
One sweep condition



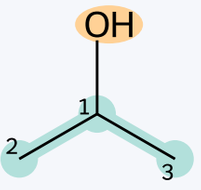
Mask one edge during growth

b Ring opening can couple an inner shuffle to an outer bond

Base: cyclopropanol



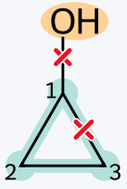
Target: propan-2-ol

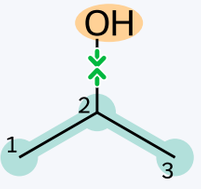


Heavy-atom events

C_1 stays next to O

1 inner + 0 outer = 1 event





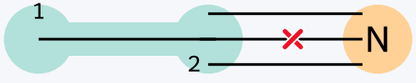
C_1 moves to a chain end

1 inner + 2 outer = 3 events

The carbon fragment is the same size; the oxygen attachment can change with its assignment.

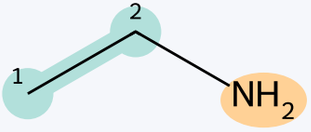
c A permissive tolerance can keep the changing bond inside a fragment

Base



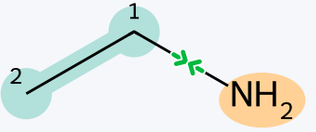
τ = 1: C–C and N separate

Target: retain carbons



1 outer weakening

Target: exchange carbons



2 outer events

τ = 2: C–N can stay inside one fragment

Heavy-atom illustrations; H constraints omitted. Red: break / weaken. Green: form / strengthen.